

Fabien Niel

Research Scientist | Former Mistral AI Applied Scientist | PhD in Theoretical Physics

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PROFILE

Research scientist with experience building and evaluating foundation-model training systems at Mistral AI and developing a production ML system from first implementation through prospective multi-hospital validation at Quantiq. PhD in theoretical physics, with expertise in statistical modelling, noisy time-series data, numerical simulation, and scientific computing.

EXPERIENCE

Mistral AI Dec. 2025 – May 2026
Applied Scientist Paris, France

- Implemented the first Mixture-of-Experts support in Forge, Mistral’s scalable framework for training and adapting frontier models, and contributed to DPO, expert parallelism, and distributed data loading.
- Ported Voxtral’s model architecture and audio data pipeline into Forge, debugging PyTorch/FSDP failures and integrating research implementations into reusable production workflows.
- Designed an evidence-grounded benchmark from raw Singaporean legal documents after the original client evaluation saturated, with controls for leakage, truthfulness, distractor plausibility, and answer-length bias.

Quantiq Feb. 2020 – Oct. 2025
Co-founder & CTO Paris, France

- Built Quantiq’s end-to-end camera-based physiological sensing system from scratch, extracting clean photoplethysmographic signals and estimating heart and respiratory rates in real time from commodity webcams and smartphone cameras under uncontrolled lighting, motion, and device conditions.
- Designed the data acquisition and evaluation infrastructure for 10,000+ participants (300+ hours of synchronized video and medical-reference signals) collected across 5+ devices, 20+ environments, and six hospitals.
- Identified severe overestimation from session-level validation by enforcing patient-disjoint splits, then introduced quality-control models and prospective clinical evaluation to measure true out-of-distribution performance.
- Developed transformer and CNN-based models for physiological time series, including self-supervised pretraining, multitask fine-tuning, architecture search, and systematic hyperparameter evaluation.
- Brought the system from first implementation to a CE-marked production medical device, leading technical direction across an eight-person team; achieved prospective RMSEs of 3.1 bpm for heart rate and 2.9 breaths/min for respiratory rate.

CNRS / École Polytechnique 2015 – 2019
PhD Researcher (2015–2018) and Postdoctoral Fellow (2018–2019) Paris, France
Visiting PhD Researcher, Max Planck Institute for Nuclear Physics Heidelberg, Germany – 2017

- Developed mathematical and numerical models of stochastic radiation reaction in strong electromagnetic fields, connecting quantum emission dynamics to classical and kinetic descriptions and establishing their domains of validity.
- Derived observable signatures of radiation reaction and strong-field QED cascades in particle and photon distributions, validating analytical predictions through numerical simulation.
- Extended this work to nonlinear Breit–Wheeler pair creation in structured laser fields; PhD thesis selected for Springer’s Outstanding PhD Thesis series.

TECHNICAL EXPERTISE

Models & methods	Transformers, GQA, RoPE, multimodal models, Mixture-of-Experts, self-supervised learning, DPO, time-series modelling
Training systems	PyTorch, FSDP, expert parallelism, distributed data pipelines, HPC and cloud GPU environments
Scientific computing	Python, numpy, numerical simulation, signal processing, Linux, Git, Docker

EDUCATION

PhD in Physics, Sorbonne Université / École Polytechnique 2015–2018
Thesis: *Classical and Quantum Description of Plasma and Radiation in Strong Fields*; Springer Outstanding PhD Thesis.

MSc in Physics, École Polytechnique – ranked 1st in the Inertial Confinement Fusion track 2014–2015

Engineering Degree, CentraleSupélec 2012–2015

SELECTED PUBLICATIONS & PATENT

- Mistral team. *Voxtral Realtime*. arXiv, 2026.
- Mistral team. *Voxtral TTS*. arXiv, 2026.
- F. Niel et al. *Prospective clinical validation of a noncontact vital signs measurement smartphone application in emergency departments*. ESC Congress, 2023.
- **Patent:** F. Niel. *System and method for obtaining a biosignal from video sequence*. EP 4 497 372 A1.
- F. Niel et al. *From quantum to classical modelling of radiation reaction: a focus on the radiation spectrum*. Plasma Physics and Controlled Fusion, 2018.
- F. Niel et al. *From quantum to classical modeling of radiation reaction: a focus on stochasticity effects*. Phys. Rev. E, 97(043209), 2018.